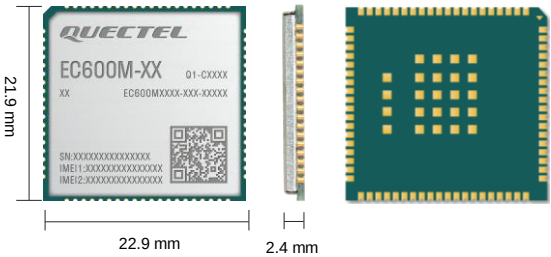


Quectel EC600M-CN

IoT/ M2M-optimized LTE Cat 1 bis Module



EC600M-CN is a cost-effective LTE Cat 1 bis wireless communication module specially designed by Quectel for M2M and IoT fields. It delivers a maximum downlink rate of 10 Mbps and a maximum uplink rate of 5 Mbps. Designed in the compact and unified form factor, EC600M-CN is compatible with LTE Standard EC600N-CN in package.

EC600M-CN adopts the laser engraving process to get a more fashionable appearance, strong metallic texture, better heat dissipation, durable label information, which makes it more suitable for automation requirements.

EC600M-CN supports interfaces such as camera, LCM, matrix keyboard, and audio to adapt to more intelligent product terminals, such as smart POS and smart handheld terminals and other high-end products.

A rich set of Internet protocols, industry-standard interfaces and abundant functionalities (USB serial drivers for Windows 8.1/ 10/ 11, Linux, Android) extend the applicability of the module to a wide range of M2M and IoT applications such as OTT, CPE, POS, data card, smart safety and industrial PDA.



Key Features

- ✓ Ultra-small size, designed for M2M and IoT applications, especially for small-size terminals
- ✓ Support remote DFOTA, reducing later operation and maintenance costs
- ✓ Support Wi-Fi Scan
- ✓ Dual SIM Single Standby
- ✓ Super cost-effective



LTE Cat 1 bis
Max. 10 Mbps (DL)
Max. 5 Mbps (UL)



LCC Package



Embedded
Abundant Protocols



USB 2.0 High Speed
Interface



USB Drivers



VoLTE (Optional)



Quectel Enhanced
AT Commands



DFOTA



Dual SIM Single Standby

Quectel EC600M-CN

LTE Cat 1 bis	EC600M-CN
Region/ Operator	China
Package	LCC
Dimensions (mm)	21.9 × 22.9 × 2.4
Weight (g)	Approx. 2.0
Temperature Range	
Operating Temperature	-35 °C to +75 °C
Extended Temperature	-40 °C to +85 °C
Frequency Bands	
LTE-FDD	B1/ 3/ 5/ 8
LTE-TDD	B34/ 38/ 39/ 40/ 41
Certifications	
Regulatory	China: SRRC/ NAL/ CCC
Others	WHQL
Data Rates	
LTE-FDD (Mbps)	10 (DL)/ 5 (UL)
LTE-TDD (Mbps)	8.96 (DL)/ 3.1 (UL)
Interfaces	
USIM ^①	× 2, 1.8/ 3.0 V
UART	× 3 (Main, Debug and Auxiliary ^②)
USB 2.0	× 1
PCM (Digital Audio) (Optional)	× 1
ADC	× 2, 12 bits
SPI (Optional) ^②	× 2
Matrix Keyboard (5 × 5) (Optional) ^②	× 1
PWM (Optional) ^②	× 4
Camera (Optional)	× 1
LCM (Optional) ^②	× 1
I2C (Optional)	× 1
Analog Audio (Optional)	Input: × 1 (Microphone) Output: × 1 (Earpiece)
Antenna	× 1
Voice (Optional)	
Speech Codec Modes	MP3/ AMR/ WAV
Voice	Digital Audio and VoLTE (Voice over LTE)
Enhanced Features	
DTMF	○
DFOTA	●
Audio Playback/ Audio Recording	○
Wi-Fi Scan	●
QuecOpen [®]	○
QuecFile [®]	●
USIM Card Detection	●
Drivers	
RIL	Android 4.x–13.x
USB RNDIS	Windows 8.1/ 10/ 11 Linux 2.6–6.7
USB ECM	Linux 2.6–6.7
USB Serial	Windows 8.1/ 10/ 11 Linux 2.6–6.7 Android 4.x–13.x
Electrical Features	
Supply Voltage Range	3.4–4.3 V, typ. 3.8 V 36 μA @ Power off 1.35 mA @ LTE-FDD Sleep (PF = 128) 1.25 mA @ LTE-FDD Sleep (PF = 256) 17.52 mA @ LTE-TDD Idle (PF = 64 USB Disconnected) 28.95 mA @ LTE-TDD Idle (PF = 64 USB Connected)

NOTE:
1. ^①: USIM2 is optional.
2. ^②: Supported only by the QuecOpen[®] solution.
3. ○: Optional.
4. ●: Supported.

